

National Sun Yat-sen University
Department of Electrical Engineering

114-2 Practical Digital System Design

HW3

Name: Enping, Chung

Student ID: B133015012

Table of Contents

PART I. Synthesis Procedure

PART II. Statistics

I. Adder

II. Voting

III. Median

PART III. Simulation Results

I. Testbench Modification

II. Waveform Analysis

PART IV. Experience and Reflection

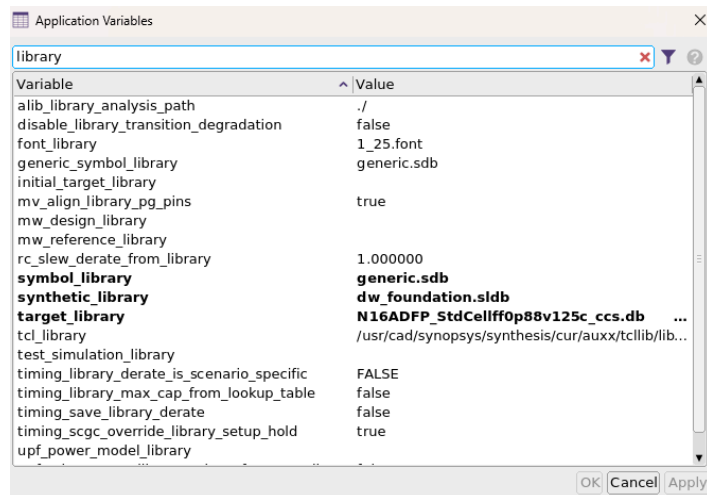
PART I. Synthesis Procedure

For the design procedure, we will be using the synthesis process of “voting.v” as the example

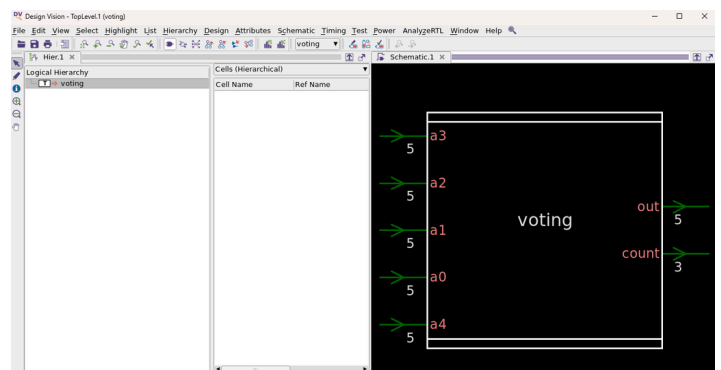
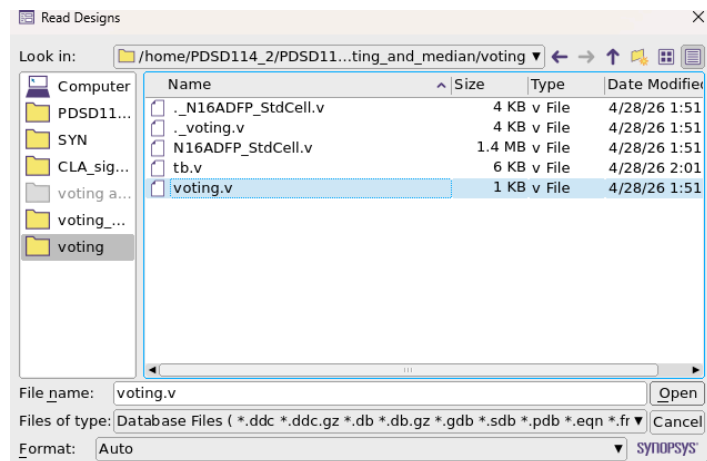
Step 1. Start Program “Design Compiler”: type dv in the terminal

```
soc09 [~/bingoHW3/attempt_2/voting_and_median/voting]
-PDSD114_2a25- $dv
```

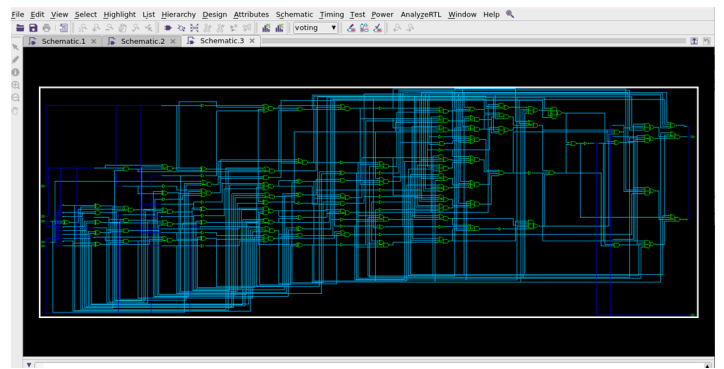
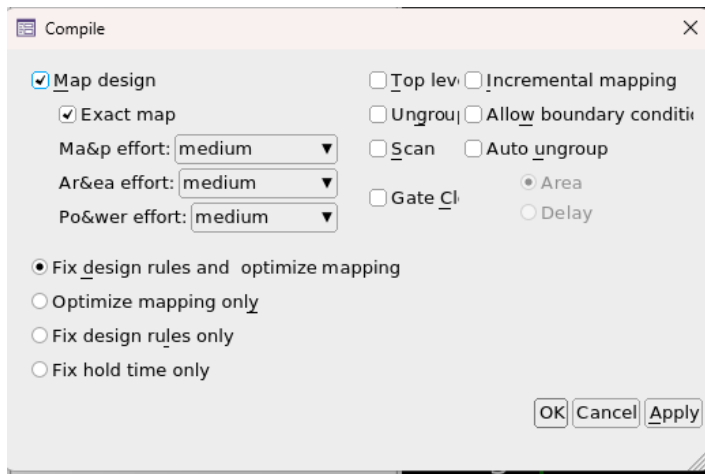
Step 2. Check and choose libraries



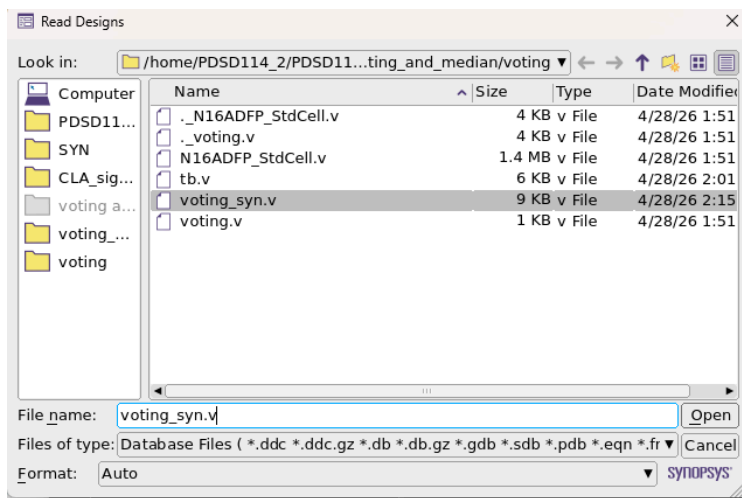
Step 3. Import(Read) File



Step 4. Compile Design



Step 5. Save Synthesized File

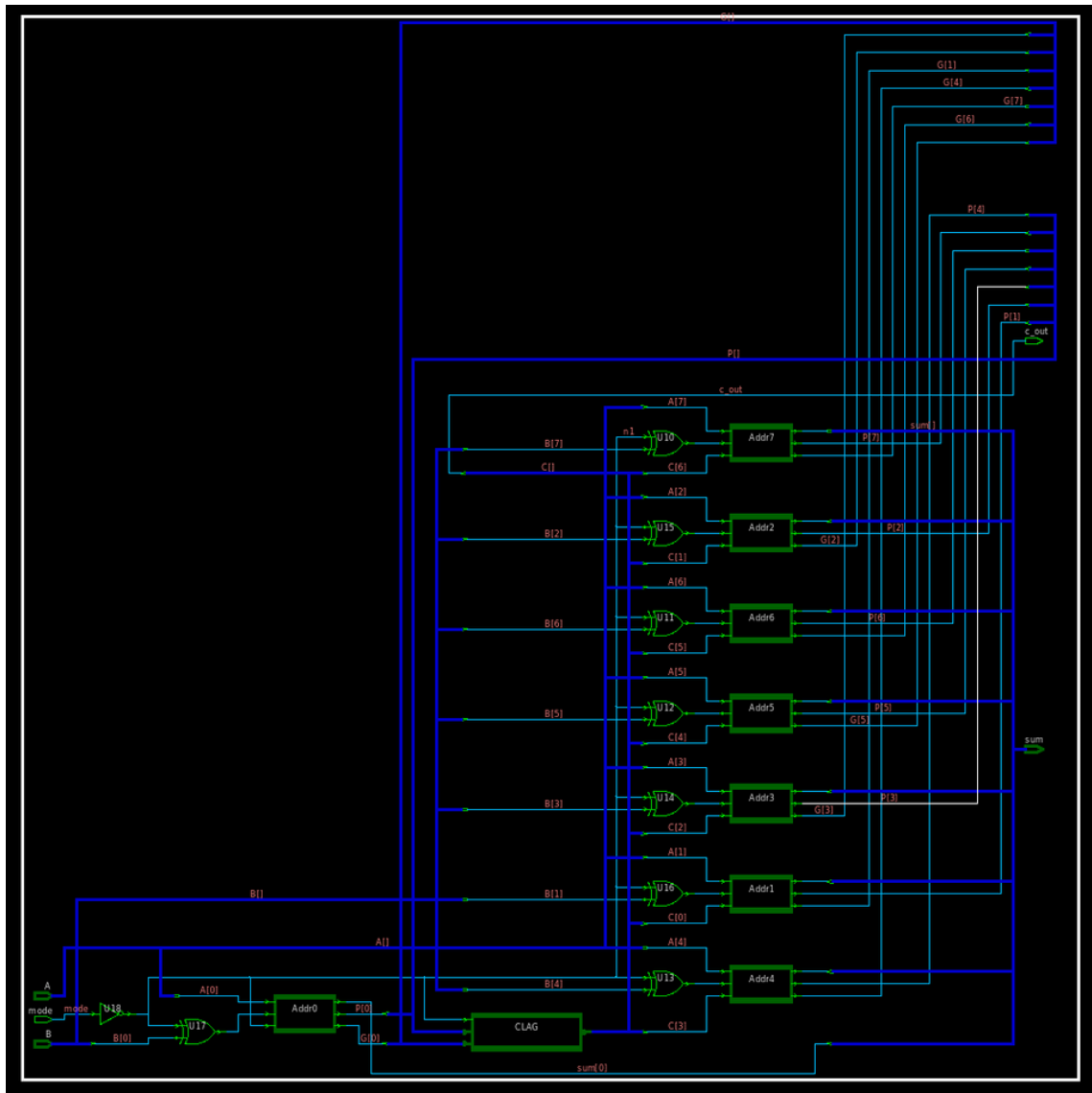


Step 6. Analyze Design (Results Shown in later Slides)

PART II. Statistics

I. Adder

Full Schematic View:



Area Report:

```

Number of ports:          99
Number of nets:          131
Number of cells:         50
Number of combinational cells: 41
Number of sequential cells: 0
Number of macros/black boxes: 0
Number of buf/inv:       1
Number of references:     11

Combinational area:      17.159040
Buf/Inv area:           0.155520
Noncombinational area:   0.000000
Macro/Black Box area:    0.000000
Net Interconnect area:   undefined (Wire load has zero net area)

Total cell area:         17.159040

```

Timing Report:

```

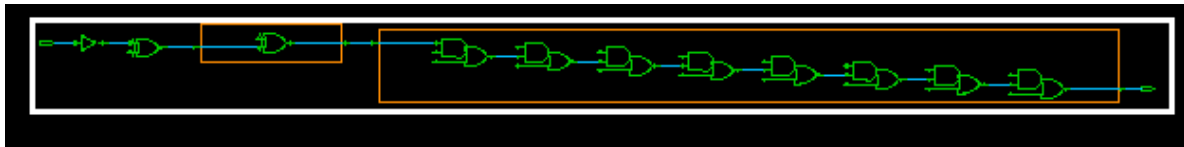
Startpoint: mode (input port)
Endpoint: c_out (output port)
Path Group: (none)
Path Type: max

```

Des/Clust/Port	Wire Load Model	Library
adder	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
eight_bit_carry_lookahead_generator	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_0	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_7	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_6	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_5	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_4	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_3	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_2	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_1	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs

Point	Incr	Path
input external delay	0.00	0.00 r
mode (in)	0.00	0.00 r
U18/ZN (CKND1BWP16P90)	0.03	0.03 f
U17/Z (XOR2D1BWP16P90)	0.02	0.05 r
Addr0/B (no_Cout_adder_0)	0.00	0.05 r
Addr0/U2/Z (XOR2D1BWP16P90)	0.02	0.07 f
Addr0/P (no_Cout_adder_0)	0.00	0.07 f
CLAG/P[0] (eight_bit_carry_lookahead_generator)	0.00	0.07 f
CLAG/U8/Z (A021D1BWP16P90)	0.02	0.09 f
CLAG/U7/Z (A021D1BWP16P90)	0.02	0.11 f
CLAG/U6/Z (A021D1BWP16P90)	0.02	0.12 f
CLAG/U5/Z (A021D1BWP16P90)	0.02	0.14 f
CLAG/U4/Z (A021D1BWP16P90)	0.02	0.16 f
CLAG/U3/Z (A021D1BWP16P90)	0.02	0.18 f
CLAG/U2/Z (A021D1BWP16P90)	0.02	0.19 f
CLAG/U1/Z (A021D1BWP16P90)	0.01	0.21 f
CLAG/C[7] (eight_bit_carry_lookahead_generator)	0.00	0.21 f
c_out (out)	0.00	0.21 f
data arrival time		0.21

Critical Path:



Critical Path:

Design	Wire Load Model	Library
adder	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_0	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
eight_bit_carry_lookahead_generator	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_1	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_2	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_3	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_4	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_5	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_6	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs
no_Cout_adder_7	ZeroWireload	N16ADFP_StdCellff0p88v125c_ccs

Global Operating Voltage = 0.88
Power-specific unit information :
Voltage Units = 1V
Capacitance Units = 1.000000pf
Time Units = 1ns
Dynamic Power Units = 1mW (derived from V,C,T units)
Leakage Power Units = 1nW

Cell Internal Power = 13.3701 uW (72%)
Net Switching Power = 5.1810 uW (28%)

Total Dynamic Power = 18.5511 uW (100%)
Cell Leakage Power = 6.5790 uW

Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

Power Group	Internal Power	Switching Power	Leakage Power	Total Power	(%)	Attrs
io_pad	0.0000	0.0000	0.0000	0.0000	(0.00%)	
memory	0.0000	0.0000	0.0000	0.0000	(0.00%)	
black_box	0.0000	0.0000	0.0000	0.0000	(0.00%)	
clock_network	0.0000	0.0000	0.0000	0.0000	(0.00%)	
register	0.0000	0.0000	0.0000	0.0000	(0.00%)	
sequential	0.0000	0.0000	0.0000	0.0000	(0.00%)	
combinational	1.3370e-02	5.1810e-03	6.5790e+03	2.5130e-02	(100.00%)	
Total	1.3370e-02 mW	5.1810e-03 mW	6.5790e+03 nW	2.5130e-02 mW		

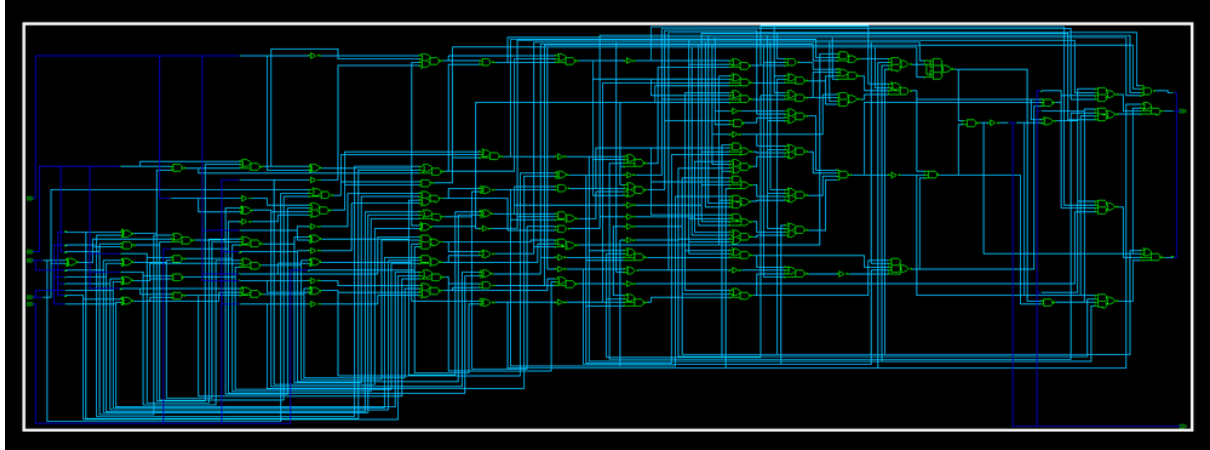
Adder – Final Design Analysis

The synthesized adder design demonstrates a balanced trade-off between area and timing performance. From the reports, the critical path is primarily dominated by the carry propagation logic, which causes delays to increase with bit width. This reflects the expected behavior of a ripple-carry structure, where higher-order bits depend on the results of lower-order carries. Although the area usage is relatively efficient, the timing performance could be improved by adopting more advanced adder

architectures such as carry-lookahead or carry-select designs. Overall, the current implementation is suitable for low-complexity applications where area efficiency is prioritized over speed.

II. Voting

Full Schematic View:



Area Report:

```
Number of ports:          33
Number of nets:           143
Number of cells:          118
Number of combinational cells: 118
Number of sequential cells: 0
Number of macros/black boxes: 0
Number of buf/inv:        26
Number of references:      26

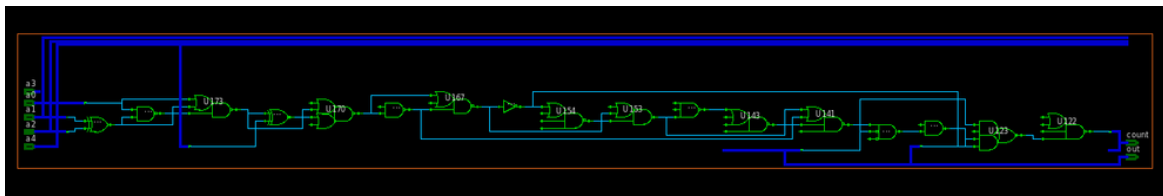
Combinational area:        34.058881
Buf/Inv area:              4.043520
Noncombinational area:     0.000000
Macro/Black Box area:      0.000000
Net Interconnect area:     undefined (Wire load has zero net area)
Total cell area:           34.058881
```

Timing Report:

Point	Incr	Path

input external delay	0.00	0.00 f
a2[0] (in)	0.00	0.00 f
U177/Z (XOR2D1BWP16P90)	0.02	0.02 r
U176/ZN (ND2D1BWP16P90)	0.01	0.03 f
U173/ZN (0AI21D1BWP16P90)	0.01	0.04 r
U172/Z (XOR2D1BWP16P90)	0.02	0.06 f
U170/ZN (0AI22D1BWP16P90)	0.01	0.07 r
U169/ZN (ND2D1BWP16P90)	0.02	0.09 f
U167/ZN (0AI21D1BWP16P90)	0.02	0.11 r
U159/ZN (CKND1BWP16P90)	0.01	0.12 f
U154/ZN (0AI211D1BWP16P90)	0.01	0.14 r
U153/Z (0A21D1BWP16P90)	0.02	0.15 r
U151/ZN (ND3D1BWP16P90)	0.01	0.16 f
U143/ZN (0AI31D1BWP16P90)	0.01	0.18 r
U141/ZN (0AI211D1BWP16P90)	0.02	0.20 f
U132/ZN (INR3D1BWP16P90)	0.01	0.21 f
U127/Z (AN2D1BWP16P90)	0.01	0.22 f
U123/ZN (A0I22D1BWP16P90)	0.01	0.23 r
U122/ZN (0AI211D1BWP16P90)	0.01	0.24 f
count[1] (out)	0.00	0.24 f
data arrival time		0.24

Critical Path:



Power Report:

Global Operating Voltage = 0.88						
Power-specific unit information :						
Voltage Units = 1V						
Capacitance Units = 1.000000pf						
Time Units = 1ns						
Dynamic Power Units = 1mW (derived from V,C,T units)						
Leakage Power Units = 1nW						
Cell Internal Power = 19.9998 uW (48%)						
Net Switching Power = 21.4280 uW (52%)						

Total Dynamic Power = 41.4278 uW (100%)						
Cell Leakage Power = 9.9089 uW						
Information: report_power power group summary does not include estimated clock tree power. (PWR-789)						
Power Group	Internal Power	Switching Power	Leakage Power	Total Power	(%)	Attrs

io_pad	0.0000	0.0000	0.0000	0.0000	(0.00%)	
memory	0.0000	0.0000	0.0000	0.0000	(0.00%)	
black_box	0.0000	0.0000	0.0000	0.0000	(0.00%)	
clock_network	0.0000	0.0000	0.0000	0.0000	(0.00%)	
register	0.0000	0.0000	0.0000	0.0000	(0.00%)	
sequential	0.0000	0.0000	0.0000	0.0000	(0.00%)	
combinational	2.0000e-02	2.1428e-02	9.9089e+03	5.1337e-02	(100.00%)	

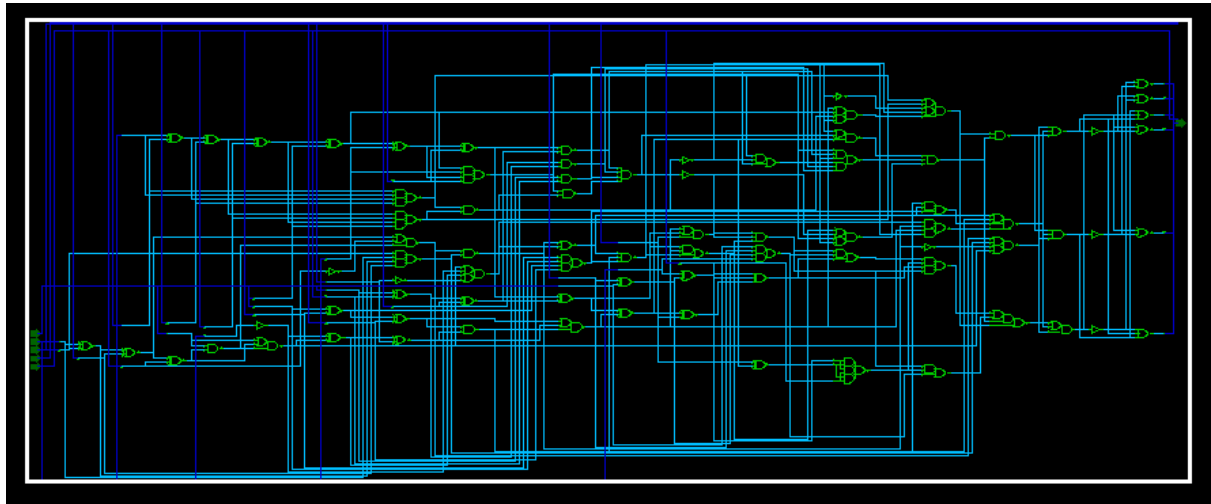
Total	2.0000e-02 mW	2.1428e-02 mW	9.9089e+03 nW	5.1337e-02 mW		

Voting – Final Design Analysis

The voting circuit shows moderate area usage due to the need to count and compare multiple inputs, but its timing performance remains relatively efficient. The critical path mainly arises from the comparison logic used to determine the maximum vote count among inputs. Since the design involves parallel counting followed by sequential comparison, the delay is influenced more by comparator depth than by arithmetic propagation. This design achieves a good balance between functionality and performance, making it suitable for applications requiring reliable majority or priority decision-making. Further optimization could focus on reducing comparator complexity to improve timing.

III. Median

Full Schematic View:



Area Report:

```
Number of ports:                36
Number of nets:                 107
Number of cells:                82
Number of combinational cells:  82
Number of sequential cells:     0
Number of macros/black boxes:   0
Number of buf/inv:              10
Number of references:           25

Combinational area:             27.578881
Buf/Inv area:                   1.555200
Noncombinational area:          0.000000
Macro/Black Box area:           0.000000
Net Interconnect area:          undefined (Wire load has zero net area)

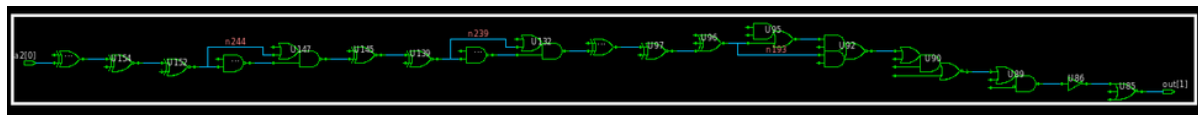
Total cell area:                27.578881
```

Timing Report:

Point	Incr	Path

input external delay	0.00	0.00 f
a2[0] (in)	0.00	0.00 f
U155/Z (XOR2D1BWP16P90)	0.02	0.02 r
U154/ZN (XNR2D1BWP16P90)	0.02	0.04 f
U152/ZN (XNR2D1BWP16P90)	0.02	0.06 r
U151/ZN (ND2D1BWP16P90)	0.01	0.07 f
U147/ZN (0AI21D1BWP16P90)	0.01	0.08 r
U145/ZN (XNR2D1BWP16P90)	0.03	0.11 f
U139/ZN (XNR2D1BWP16P90)	0.02	0.12 r
U138/ZN (ND2D1BWP16P90)	0.02	0.14 f
U132/Z (0A21D1BWP16P90)	0.02	0.16 f
U117/Z (XOR2D1BWP16P90)	0.03	0.18 r
U97/ZN (XNR2D1BWP16P90)	0.02	0.20 f
U96/Z (XOR3D1BWP16P90)	0.03	0.23 r
U95/Z (A021D1BWP16P90)	0.01	0.24 r
U92/ZN (A0I22D1BWP16P90)	0.01	0.25 f
U90/ZN (0A0I211D1BWP16P90)	0.01	0.26 r
U89/ZN (IA021D1BWP16P90)	0.02	0.28 r
U86/ZN (CKND1BWP16P90)	0.01	0.29 f
U85/ZN (NR3D1BWP16P90)	0.01	0.30 r
out[1] (out)	0.00	0.30 r
data arrival time		0.30

Critical Path:



Power Report:

Global Operating Voltage = 0.88					
Power-specific unit information :					
Voltage Units = 1V					
Capacitance Units = 1.000000pf					
Time Units = 1ns					
Dynamic Power Units = 1mW (derived from V,C,T units)					
Leakage Power Units = 1nW					
Cell Internal Power = 23.6485 uW (59%)					
Net Switching Power = 16.3982 uW (41%)					

Total Dynamic Power = 40.0467 uW (100%)					
Cell Leakage Power = 8.0062 uW					
Information: report_power power group summary does not include estimated clock tree power. (PWR-789)					
Power Group	Internal Power	Switching Power	Leakage Power	Total Power	(%) Attrs

io_pad	0.0000	0.0000	0.0000	0.0000	(0.00%)
memory	0.0000	0.0000	0.0000	0.0000	(0.00%)
black_box	0.0000	0.0000	0.0000	0.0000	(0.00%)
clock_network	0.0000	0.0000	0.0000	0.0000	(0.00%)
register	0.0000	0.0000	0.0000	0.0000	(0.00%)
sequential	0.0000	0.0000	0.0000	0.0000	(0.00%)
combinational	2.3649e-02	1.6398e-02	8.0062e+03	4.8053e-02	(100.00%)

Total	2.3649e-02 mW	1.6398e-02 mW	8.0062e+03 nW	4.8053e-02 mW	

Median – Final Design Analysis

The median design exhibits higher complexity compared to the adder and voting circuits, resulting in increased area and longer critical path delay. This is due to the need for multiple comparisons and conditional selections to correctly determine the middle value. The critical path is dominated by cascaded comparison operations, which create deeper logic levels. While the design ensures correct functionality, its performance may become a limitation in timing-sensitive applications. Optimizations such as parallel comparison structures or more efficient sorting networks could be explored to reduce delay. Overall, the design highlights the trade-off between computational complexity and hardware efficiency in implementing decision-based logic.

PART III. Simulation Results

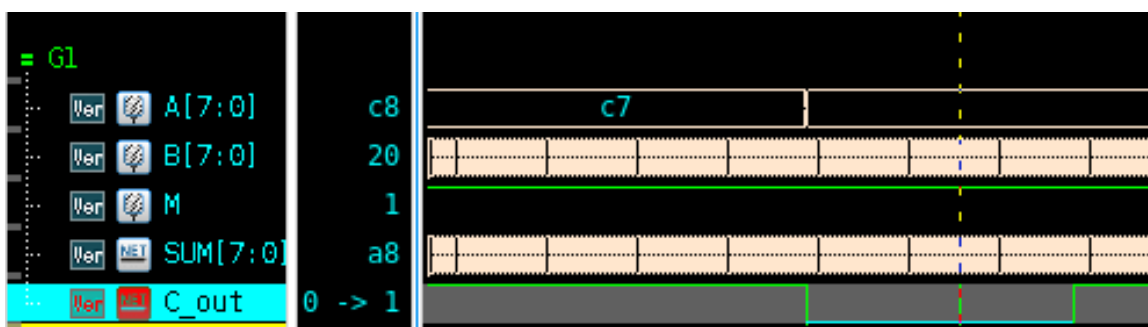
I. Testbench Modification

Before doing the simulation again, we need to make some modifications to the test bench. First, the hold (HOLD_NS) and settle(SETTLE_NS) time is extended to account for more real world accuracy. Next, the architecture file is included into the testbench. Finally, we add a additional line to write to the .sdf(standard delay format) file.

Terminal Output(using adder as emaxple):

```
Testing.....
[ALL CASES]-----
ALL SUMMARY: PASS (131072 tests)
[SELECTED CASES]-----
===== testbench: relevant functional + edge-case vectors =====
| time(ns) | M | A | B | DUT_SUM | DUT_C | EXP_SUM | EXP_C | RESULT | Description |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 3407912000 | 1 | 12 | 34 | 46 | 0 | 46 | 0 | PASS | ADD basic: 0x12 + 0x34 |
| 3407930000 | 1 | ff | 01 | 00 | 1 | 00 | 1 | PASS | ADD carry-out: 0xFF + 0x01 |
| 3407948000 | 1 | ff | ff | fe | 1 | fe | 1 | PASS | ADD max + max: 0xFF + 0xFF |
| 3407966000 | 1 | 00 | 00 | 00 | 0 | 00 | 0 | PASS | ADD zero + zero |
| 3407984000 | 0 | 3a | 19 | 21 | 1 | 21 | 1 | PASS | SUB basic: 0x3A - 0x19 |
| 3408002000 | 0 | 05 | 03 | 02 | 1 | 02 | 1 | PASS | SUB no borrow: 0x05 - 0x03 |
| 3408020000 | 0 | 00 | 01 | ff | 0 | ff | 0 | PASS | SUB borrow: 0x00 - 0x01 |
| 3408038000 | 0 | 00 | ff | 01 | 0 | 01 | 0 | PASS | SUB underflow edge: 0x00 - 0xFF |
| 3408056000 | 0 | a5 | a5 | 00 | 1 | 00 | 1 | PASS | SUB self: A - A |
| 3408074000 | 1 | a5 | 00 | a5 | 0 | a5 | 0 | PASS | ADD identity: A + 0 |
| 3408092000 | 0 | a5 | 00 | a5 | 1 | a5 | 1 | PASS | SUB identity: A - 0 |
| 3408110000 | 1 | 55 | aa | ff | 0 | ff | 0 | PASS | ADD alternating bits: 0x55 + 0xAA |
| 3408128000 | 0 | aa | 55 | 55 | 1 | 55 | 1 | PASS | SUB alternating bits: 0xAA - 0x55 |
| 3408146000 | 1 | 0f | 01 | 10 | 0 | 10 | 0 | PASS | ADD nibble carry chain: 0x0F + 0x01 |
===== SUMMARY =====
Total checks: 14
Errors: 0
RESULT: PASS
```

II. Waveform Analysis



Using Adder as Example

The main Difference is here, where you can see the C_out takes time to propagate results, if this was a pre-synthesized file without architechure, there will be no delay unless explicitly written in the code.

What is the Delay Effect?

In general, the delay effect in digital circuits means signals need a finite time to travel through logic gates and wires, so outputs respond after a propagation delay instead of changing instantly. At RTL, logic is usually idealized as zero delay, but in gate-level simulation with timing annotation, every gate contributes real delay, so different paths arrive at different times and temporary glitches can appear before the final stable value. Using your adder as an example, when ``A``, ``B``, or ``mode`` changes, the result is computed through multiple gates (invert/XOR for add-sub control, propagate/generate logic, carry network, then sum XORs), so lower bits may settle earlier while upper bits and ``c_out`` settle later due to longer carry-related paths. This staggered settling and possible short transient wrong values are exactly the delay effect you look for in the waveform.

PART IV. Experience and Reflection

I think the fun part of this assignment is learning how to use the synthesis tools. At first, the process of setting up libraries, compiling designs, and analyzing reports felt complicated, but after practicing, I became more familiar with the workflow and its purpose in real-world circuit design.

A problem I met was that during my first attempt, I named my folder “attempt 1” which apparently was not allowed since now my Design Compiler has now problems reading libraries properly. The easy fix was to change the name to “attempt_1”. The reason for this is that a lot of tools in Linux are built on older UNIX systems where spaces are treated as separators.

An annoying part was that the Design Compiler Crashed a few times during the process (less compared to when working on previous assignments due to some circumstances).

In terms of timing analysis, I observed that some designs had relatively long critical paths, especially in more complex circuits like the adder and median modules. These delays were mainly caused by cascaded logic operations and comparison circuits. To address this, I analyzed the timing reports to identify the sources of delay and considered possible architectural improvements, such as reducing logic depth or restructuring the design. This helped me understand that achieving good performance requires not only functional correctness but also careful design optimization.

Another important takeaway was observing the differences between RTL simulation and gate-level simulation. Before this assignment, I mainly worked with ideal simulations where signals changed instantly. However, by including timing information and analyzing waveforms, I was able to clearly see propagation delays and understand how signals actually behave in hardware. This helped me better appreciate concepts like critical path and timing constraints, which are essential for ensuring correct circuit operation.

I also took time to redo my earlier assignments where I had a lot of points deducted (wrong module functions and better testbench designs)

Overall, this assignment strengthened both my theoretical understanding and practical skills in digital system design. It bridged the gap between coding and physical implementation, and helped me build confidence in using professional design tools.